



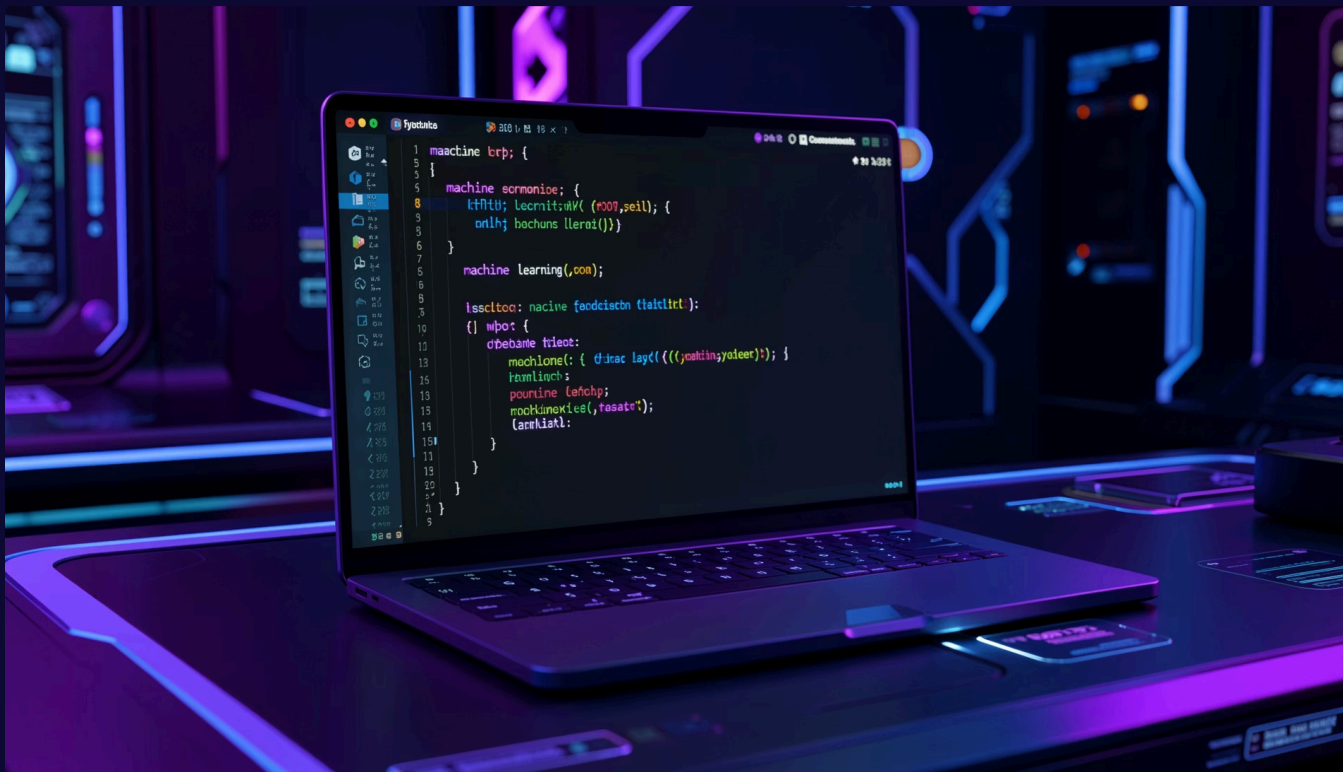
Pomegranate Disease Detection

A CNN-based software prototype for intelligent fruit disease classification
— designed for real-world deployment on embedded hardware.

FINAL YEAR PROJECT

COMPUTER VISION · DEEP LEARNING

Why a Software-Based Prototype?



Strategic Design Decision

Due to hardware procurement constraints, we implemented a fully functional **software-based prototype** — a recognized and academically valid methodology in embedded systems research.

- i "The system is architected to run on Raspberry Pi with a live camera. Currently, we demonstrate the core intelligence layer using a software prototype — the hardware integration is modular and can be added at any stage."

System Architecture



The architecture cleanly separates the intelligence layer from the hardware layer — making the CNN model portable, testable, and hardware-agnostic by design.



Dataset-Driven Development

We sourced labeled pomegranate fruit images from **Kaggle's Pomegranate Disease Dataset** — replacing live camera input with a curated, high-quality image corpus.

More Training Data

Hundreds of labeled samples
vs. a handful of live captures

Better Generalization

Diverse lighting, angles, and
disease stages included

Reproducible Results

Consistent benchmark for model evaluation and comparison

CNN Model Workflow

01

Load Dataset

Read pomegranate images from organized class folders
(Healthy / Diseased)

02

Preprocess & Augment

Resize, normalize, and augment images for robust training

03

Train CNN Model

Feed data through convolutional layers to extract disease
features

04

Test & Predict

Run sample images through the trained model and output
classification

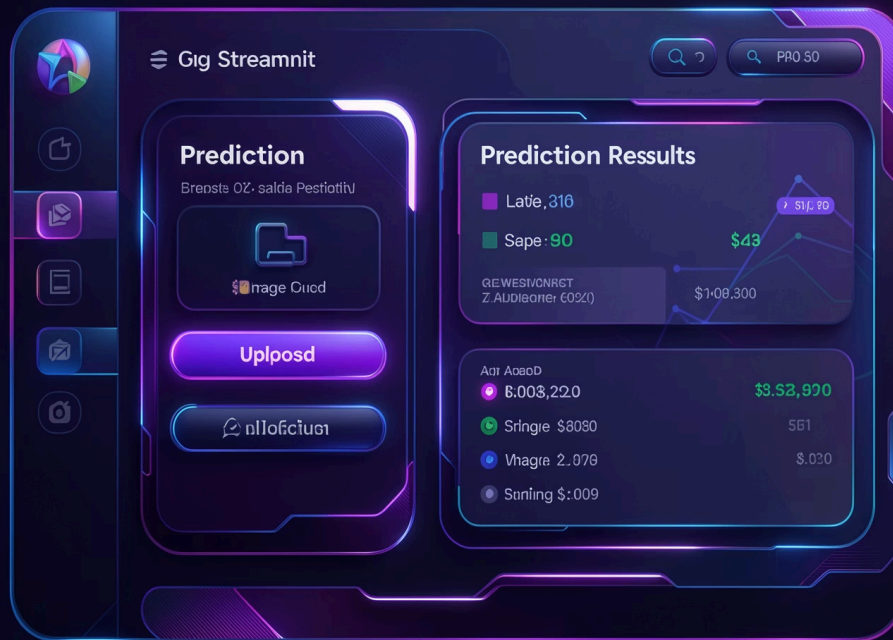
Demo Interface

Built with Python (Streamlit)

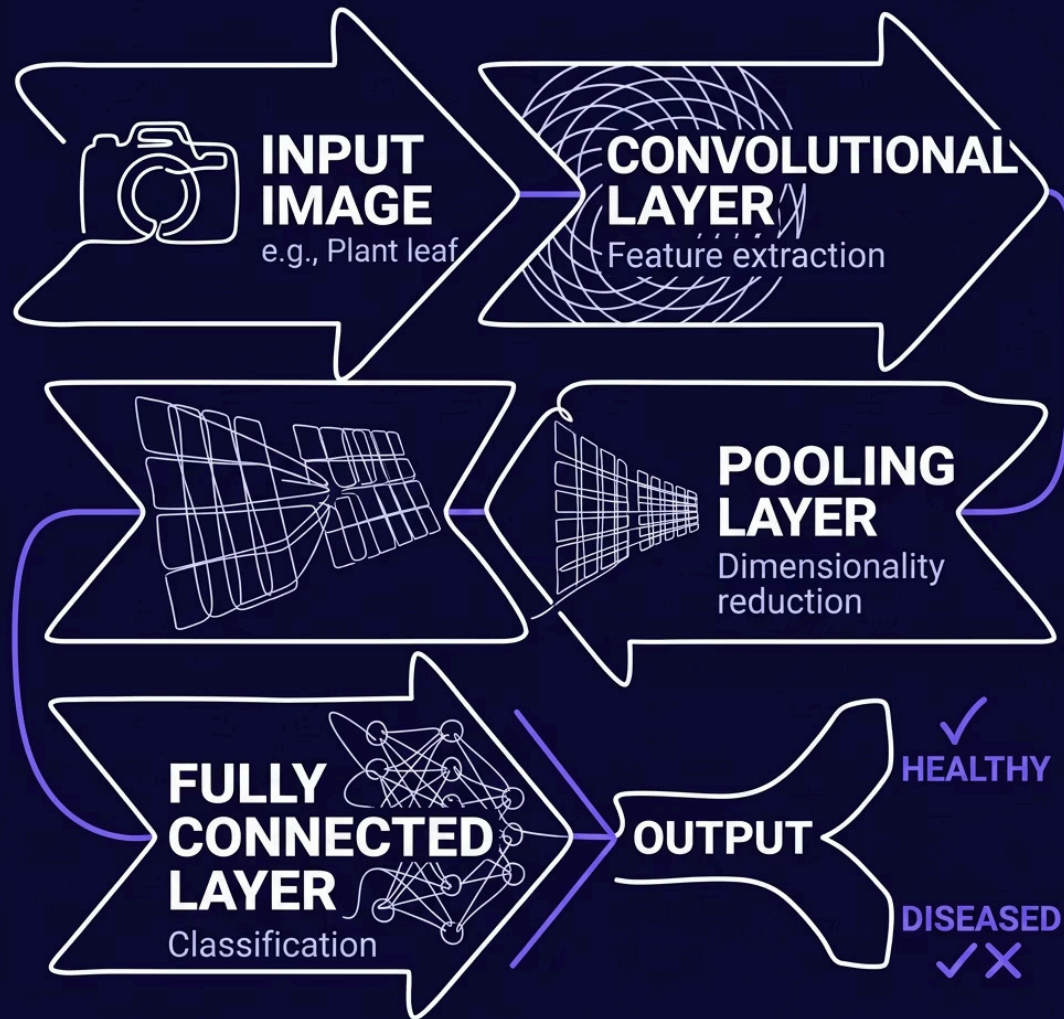
A clean, interactive demo interface that makes the model accessible and evaluable without any command-line interaction.

- **Upload** a pomegranate image
- Click "**Predict**" to run inference
- View result: **Healthy** or **Diseased** with confidence score

- ✓ The interface simulates exactly what a farmer-facing app would look like in field deployment.



Core Intelligence: The CNN Model

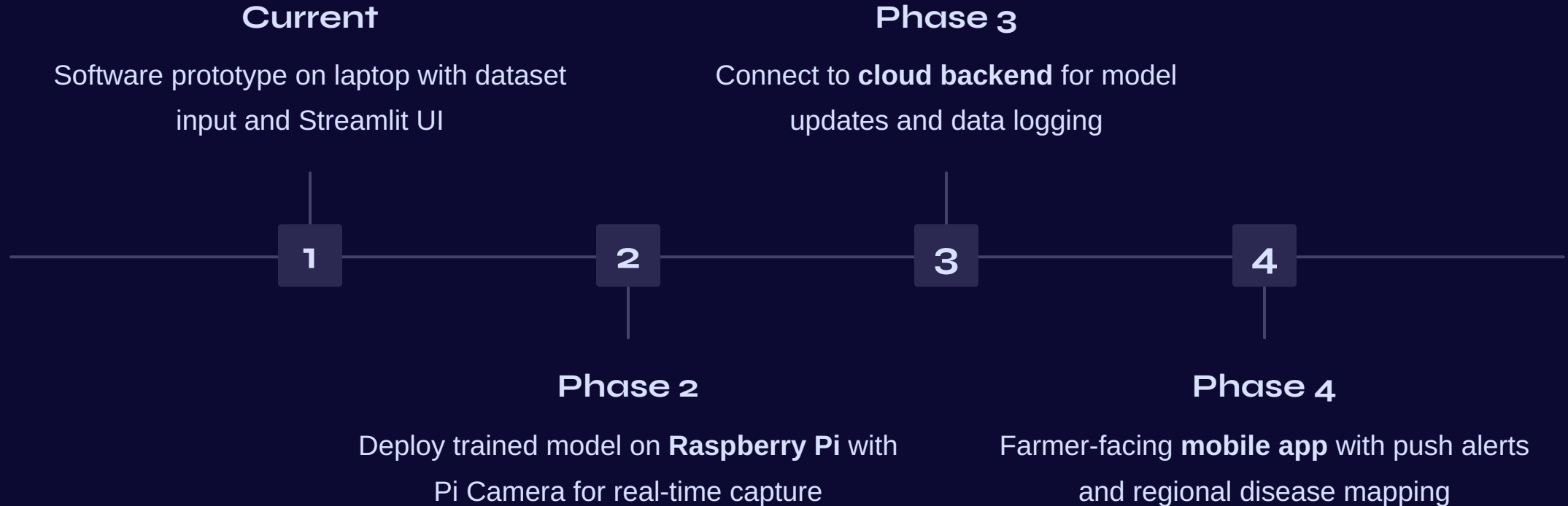


What the CNN Learns

The convolutional layers automatically detect visual disease markers — discoloration, lesions, texture anomalies — without manual feature engineering.

Key design principle: The intelligence of detection is fully hardware-independent. The same trained model runs on a laptop today and Raspberry Pi tomorrow.

Future Scope: Hardware Deployment



The modular architecture ensures each phase is independently deliverable — hardware integration requires **no changes** to the core model.

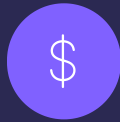
Real-World Impact

Early disease detection can prevent **significant crop losses** and reduce pesticide overuse. This system brings that capability to any farmer with a smartphone camera.



Early Detection

Catch disease before it spreads across the harvest



Reduced Losses

Targeted intervention instead of blanket pesticide application



Accessible Tool

No agricultural expertise required — just point and predict



Key Takeaways

Software Prototype = Valid Methodology

Dataset-driven training yields **more accurate, reproducible results** than limited live-camera data — a deliberate and defensible design choice.

Intelligence is Modular

"Our focus is on the intelligence of detection using CNN.
Hardware integration is modular and can be added at any time."

Production-Ready Architecture

The pipeline — Input → CNN → Prediction → Output — maps directly onto the Raspberry Pi deployment path with no architectural changes required.